

Francisco Teixeira

Postdoctoral Researcher

Email | Github | LinkedIn | Scholar

Postdoctoral researcher working on deep learning systems for speech processing. My main expertise spans privacy-preserving machine learning – including voice conversion and anonymization, membership inference, machine unlearning and cryptographic processing – speech recognition, speaker recognition and speaker diarization. Recently co-developed CAMÕES, an evaluation benchmark and set of state-of-the-art open-source ASR models for European Portuguese, and led the collection of FaLAR, a 5,000-hour speaker-annotated speech corpus (available on HuggingFace). I have also recently participated in the development of SVeritas, a robustness benchmark for automatic speaker verification. I am currently the Principal Investigator of the LeaF project (CMU-Portugal), focused on machine unlearning in speech foundation models. I have a PhD from INESC-ID/Instituto Superior Técnico, University of Lisbon, Portugal, obtained in 2024, with a visiting research period at the Language Technologies Institute at Carnegie Mellon University.

Current Activities

- Postdoctoral Researcher at the Human Language Technologies (HLT) Lab under the *ACCELERAT.AI* (PRR) project at INESC-ID, Lisbon, Portugal (2024–present).
- Principal Investigator of the *LeaF* exploratory research project on machine unlearning in speech foundation models (2025–present).

Technical Skills

■ Deep Learning & Speech Processing

- PyTorch, HuggingFace Transformers, ESPnet, SpeechBrain, Pyannote, Kaldi;
- Experience with speech foundation models: Whisper, wav2vec 2.0, HuBERT, WavLM;
- Membership inference, adversarial robustness, machine unlearning.

■ Privacy-Preserving ML

- Speaker anonymization, voice conversion, information disentanglement.
- Homomorphic encryption (SEAL, OpenFHE), secure multi-party computation (ABY, MP-SPDZ);

■ Programming & General

- Python (primary), C++, Java, C, MATLAB;
- Basic: MySQL, HTML, PHP, JavaScript.

Languages: Fluency in English, B1-level Italian, basic understanding of Spanish and French.

Recent achievements

- Collection and curation of [FaLAR](#), the first open-source large-scale European Portuguese Speech corpus, with over 5,000 hours of speaker-annotated data.
- Co-development of [CAMÕES](#), the current state-of-the-art open-source model for European Portuguese ASR, and its corresponding evaluation benchmark.
- Participation in the development of the [SVeritas](#) benchmark for robust speaker verification.

Education & Research

■ Academic Qualifications

- 2019–2024** Doctoral Degree in Electrical and Computer Engineering
Instituto Superior Tecnico, University of Lisbon, Portugal
PhD Thesis: [Privacy-preserving Machine Learning in Remote Speech Processing](#)
Qualification: Pass with Distinction and Honour
- 2013–2018** Integrated Master's in Electrical and Computer Engineering
Instituto Superior Tecnico, University of Lisbon, Portugal
MSc Thesis: [Privacy in Paralinguistic Tasks](#)
- 2012–2013** 1st Year of Bachelor in Information Systems and Computer Engineering
Instituto Superior Tecnico, University of Lisbon, Portugal
- February – July 2017** Erasmus Program student
University of Siena, Siena, Italy

■ Research

- March – June 2022** Visiting Researcher
Language Technologies Institute, Carnegie Mellon University, Pittsburgh, USA
- October 2018 – February 2019** Research Intern
Human Language Technology (HLT) Lab, INESC-ID, Lisbon, Portugal

Selected publications

- 2026** Exploring the potential and limitations of Model Merging for Multi-Domain Adaptation in ASR
Carvalho, C., **Teixeira, F.**, Rolland, T. and Abad, A. – Accepted for publication in *Interspeech 2026*
- 2026** FaLAR: A Large-scale Speaker-Annotated European Portuguese Speech Corpus of Parliamentary Sessions
Teixeira, F., Carvalho, C., Julião, M., Botelho, C., Solera-Ureña, R., Paulo, S., Rolland, T., Peters, B., Trancoso, I. and Abad, A. – Published in *LREC 2026*
- 2025** CAMÕES: A Comprehensive Automatic Speech Recognition Benchmark for European Portuguese
Carvalho, C., **Teixeira, F.**, Botelho, C., Pompili, A., Solera-Ureña, R., Paulo, S., Julião, M., Rolland, T., Mendonca, J., Pereira, D., Trancoso, I., Abad, A. – Published in *ASRU 2025*
- 2025** SVeritas: Benchmark for Robust Speaker Verification under Diverse Conditions
Baali, M., Bisht, S., **Teixeira, F.**, Shapovalenko, K., Singh, R., Raj, B. – Published in *EMNLP Findings 2025*
- 2025** Children's Voice Privacy: First Steps And Emerging Challenges
Kulkarni, A., **Teixeira, F.**, Hermann, E., Rolland, T., Trancoso, I., Magimai.-Doss, M. – Published in *Interspeech 2025*
- 2025** Exploring Features for Membership Inference in ASR Model Auditing
Teixeira, F., Pizzi, K., Olivier, R., Abad, A., Raj, B., Trancoso, I. – Published in *Computer Speech & Language – Journal*
- 2024** Privacy-oriented Manipulation of Speaker Representations
Teixeira, F., Abad, A., Raj, B., Trancoso, I. – Published in *IEEE Access 2024*
- 2022** Towards End-to-End Private Automatic Speaker Recognition
Teixeira, F., Abad, A., Raj, B., Trancoso, I. – Published in *Interspeech 2022*

Projects

Machine Unlearning in Speech Foundation Models – Learning to Forget (LeaF)
Principal Investigator, CMU Portugal Exploratory Research Project, 2025/11 –

Privacy for Smart Speech Technology (PSST!)
Researcher, Horizon Europe Marie Skłodowska-Curie Action, 2025–

ACCELERAT.AI
Researcher, Portuguese Recovery and Resilience Plan (PRR), 2024–

Privacy in speaker diarization: secure detection of who spoke when (PrivaDia)
Researcher, CMU Portugal Exploratory Research Project, 2021/01–2022/06

Teaching

■ Courses

Foundations of Programming
Teaching Assistant, Instituto Superior Tecnico, University of Lisbon, 1st semester, 2025/2026

Speech Processing
Teaching Assistant, Instituto Superior Tecnico, University of Lisbon, 2nd semester, 2018/2019

■ Supervisions

Voice Anonymization for European Portuguese
MSc. thesis, 2025–(ongoing)

Multimodal Anonymization for European Portuguese
MSc. thesis, 2025–(ongoing)

Machine Unlearning against Jailbreak Attacks in Speech Language Models
MSc. thesis, 2025–(ongoing)

Talks

[POSTER] Privacy-preserving Machine Learning for Remote Speech Processing
IberSPEECH, November 2024

[ORAL] Improving Membership Inference in ASR Model Auditing with Perturbed Loss Features
Trustworthy Speech Processing Workshop – ICASSP, April 2024

[ORAL] Privacy-preserving Machine Learning for Speech
Invited Lecture, Natural Language Processing course, ISCTE, Lisbon, November 2023

[ORAL] Privacy-Preserving Automatic Speaker Diarization
ICASSP, June 2023

[POSTER] Privacy in Automatic Speaker Verification and Diarization
CMU-Portugal Summit, October 2022

[WEBINAR] Towards End-to-End Private Automatic Speaker Recognition
ISCA SIG-SPSC, 2022

[ORAL] Towards End-to-End Private Automatic Speaker Recognition
Interspeech, September 2022

[ORAL] FoolHD - Fooling speaker identification by Highly imperceptible adversarial Disturbances
ICASSP, June 2021

[WEBINAR] Privacy in Health Oriented Paralinguistic and Extralinguistic Tasks
ISCA SIG-SPSC, July 2020

[POSTER] Privacy-preserving Paralinguistic Tasks
ICASSP, May 2019

Other activities

- Workshop for Young Female Researchers in Speech Science & Technology – Program Committee member, 2021, 2022.
- Participant at Dagstuhl Seminar on Privacy in Speech and Language Technology, August 2022.
- Participant at the Lorentz Center Workshop on Speech as Personal Identifiable Information, November 2021.
- Monitor at the 11th Lisbon Machine Learning School (LxMLS), July 2021.
- ISCA-SAC's Board Advisor, 2023.
- ISCA-SAC's Event Coordinator, 2022.
- ISCA-SAC's Media Coordinator, 2021.
- ISCA-SAC Student volunteer at Interspeech, October 2020.
- ISCA-SAC's SpeechPitch podcast co-founder and co-host, 2021–2023.
- Participant at the 9th Lisbon Machine Learning School (LxMLS), July 2019.
- Volunteer co-organizer of the 3rd Annual MC Meeting of the enetCollect COST Action (CA16105), March 2019.
- Participant at the TAPAS TE2: Speech Processing and Machine Learning Workshop, February 2019.
- Team supervisor at GASTagus ONG (Tagus Group for Social Action) – 2016.
- Volunteer at GASTagus ONG (Tagus Group for Social Action) – 2015.

Reviewing

2026	Reviewer Interspeech
2026	Reviewer ICASSP
2025	Reviewer Disfluency in Spontaneous Speech (DiSS) Workshop
2025	Reviewer Computer Speech & Language – Journal
2025	Reviewer Interspeech
2024	Reviewer Proceedings IEEE – Journal
2023	Reviewer IEEE Transactions on Information Forensics & Security (T-IFS)
2023	Reviewer ACM International Conference on Multimodal Interaction (ICMI)
2020	Reviewer IEEE International Conference on Machine Learning and Applications (ICMLA)
2020	Secondary reviewer <i>Neural Networks</i> – Journal
2020	Secondary reviewer International Conference on Statistical Language and Speech Processing (SLSP)

Full list of publications

- 2026 Evaluating Parkinson's Disease Detection in Anonymized Speech: A Performance and Acoustic Analysis
Franzreb, C., **Teixeira, F.**, Luks, B. Möller, S. and Abad, A. – Accepted for publication in *Symposium on Security and Privacy in Speech Communication (SPSC) 2026*
- 2026 ESPnet3: Infrastructure for Scalable Speech and Audio Research in the Foundation Model Era
Someki, M., Polok, A., Carvalho, C., Lin, C., Yang, D., Shi, J., Tian, J., Soplin, N., Cornell, S., Arora, S., **Teixeira, F.**, Wang, W., Chen, W., Abad, A., Li, C., Watanabe, S., Zhang, W. – Accepted for publication in *Interspeech 2026*
- 2026 Exploring the potential and limitations of Model Merging for Multi-Domain Adaptation in ASR
Carvalho, C., **Teixeira, F.**, Rolland, T. and Abad, A. – Accepted for publication in *Interspeech 2026*
- 2026 FaLAR: A Large-scale Speaker-Annotated European Portuguese Speech Corpus of Parliamentary Sessions
Teixeira, F., Carvalho, C., Julião, M., Botelho, C., Solera-Ureña, R., Paulo, S., Rolland, T., Peters, B., Trancoso, I. and Abad, A. – Published in *LREC 2026*
- 2025 SVeritas: Benchmark for Robust Speaker Verification under Diverse Conditions
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- 2025 CAMÕES: A Comprehensive Automatic Speech Recognition Benchmark for European Portuguese
Carvalho, C., **Teixeira, F.**, Botelho, C., Pompili, A., Solera-Ureña, R., Paulo, S., Julião, M., Rolland, T., Mendonca, J., Pereira, D., Trancoso, I., Abad, A. – Published in *ASRU 2025*
- 2025 Children's Voice Privacy: First Steps And Emerging Challenges
Kulkarni, A., **Teixeira, F.**, Hermann, E., Rolland, T., Trancoso, I., Magimai.-Doss, M. – Published in *Interspeech 2025*
- 2025 Tackling Cognitive Impairment Detection from Speech: A submission to the PROCESS Challenge
Botelho, C., Gimeno-Gomez, D., **Teixeira, F.**, Mendonca, J., Pereira, P., AP Nunes, D., Rolland, T., Pompili, A., Solera-Ureña, R., Ponte, M., Martins de Matos, D., Martinez-Hinarejos, C., Trancoso, I., Abad, A. – Published in *Interspeech 2025*
- 2025 Exploring Features for Membership Inference in Asr Model Auditing
Teixeira, F., Pizzi, K., Olivier, R., Abad, A., Raj, B., Trancoso, I. – Published in *Computer Speech and Language*
- 2024 Privacy-oriented Manipulation of Speaker Representations
Teixeira, F., Abad, A., Raj, B., Trancoso, I. – Published in *IEEE Access 2024*
- 2024 Improving Membership Inference in ASR Model Auditing using Loss-based Features
Teixeira, F., Pizzi, K., Olivier, R., Abad, A., Raj, B., Trancoso, I. – Published in Trustworthy Speech Processing Workshop, *ICASSP 2024*
- 2023 Privacy-preserving Automatic Speaker Diarization
Teixeira, F., Abad, A., Raj, B., Trancoso, I. – Published in *ICASSP 2023*
- 2023 Treating Speech as Personally Identifiable Information and Its Impact in Machine Translation
Trancoso, I., **Teixeira, F.**, Botelho, C., Abad, A. In Moniz, H., Parra Escartin, C. (eds) Towards Responsible Machine Translation. Machine Translation: Technologies and Applications, vol 4. Springer, Cham.
- 2022 Towards End-to-End Private Automatic Speaker Recognition
Teixeira, F., Abad, A., Raj, B., Trancoso, I. – Published in *Interspeech 2022*

- 2021** Voice biometrics: privacy in paralinguistic and extralinguistic tasks for health applications
Teixeira, F., Abad, A., Trancoso, I., Raj, B. – Published in IET Book chapter 4 of Voice Biometrics: Technology, trust and security;
 Eds. by Garcia-Mateo, C. and Chollet, G., 2021, ISBN: 978-1-78561-900-7
- 2021** Transfer Learning-Based Cough Representations for Automatic Detection of COVID-19
 Solera, R., Botelho, C., **Teixeira, F.**, Rolland, T., Abad, A., Trancoso, I. – Published in *Interspeech* 2021
- 2021** FoolHD - Fooling speaker identification by Highly imperceptible adversarial Disturbances
 Shamsabadi, A., **Teixeira, F.**, Abad, A., Raj, B., Cavallaro, A., Trancoso, I. – Published in *ICASSP* 2021
- 2021** The in-the-Wild Speech Medical Corpus
 Correia, J., **Teixeira, F.**, Botelho, C., Trancoso, I., Raj, B. – Published in *ICASSP* 2021
- 2020** Analyzing Breath Signals for the Interspeech 2020 ComParE Challenge
 Mendonca, J., **Teixeira, F.**, Abad, A., Trancoso, I. – Published in *Interspeech* 2020
- 2019** Preserving Privacy in Speaker and Speech Characterisation
 Nautsch, A., Jimenez, A., Treiber, A., Kolberg, J., Jasserand, C., Kindt, E., Delgado, H., Todisco, M., Hmani, M. A., Mtibaa, A., Abdelraheem, M. A., Abad, A., **Teixeira, F.**, Matrouf, D., Gomez-Barrero, M., Petrovska-Delacretaz, D., Chollet, G., Evans, N., Schneider, T., Bonastre, J.F., Raj, B., Trancoso, I., Busch, C. – Published in *Computer Speech & Language*
- 2019** Privacy-preserving Paralinguistic Tasks
Teixeira, F., Abad, A., Trancoso, I. – Published in *ICASSP* 2019
- 2018** Analysing speech for clinical applications
 Trancoso, I., Correia, J., **Teixeira, F.**, Raj, B., Abad, A. – Keynote Talk at *SLSP*
- 2018** Speech analytics for medical applications
 Trancoso, I., Correia, J., **Teixeira, F.**, Raj, B., Abad, A. – Keynote Talk at *TSD*
- 2018** Mining multimodal repositories for speech affecting diseases
 Correia, J., Raj, B., Trancoso I., **Teixeira, F.** – Published in *Interspeech* 2018
- 2018** Patient Privacy in Paralinguistic Tasks
Teixeira, F., Abad, A., Trancoso, I. – Published in *Interspeech* 2018